A Multi-flow-driven Mechanism to Support Live Video Streaming on VANETs

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Abstract. *Vehicular Ad Hoc Networks (VANETs) promise a wide scope of real-time multimedia services, which need to deal with strict quality level requirements and highly dynamic network topology. To handle these challenges, multi-path routing approaches have been applied to bring improvements in Quality of Experience (QoE) levels and support on-road real-time video delivery. Though, in real situations, there will likely be multiple streams simultaneously, which can cause more congestion periods and packet loss rates. This paper proposes the Multi-flow-driven VIdeo DElivery (MVIDE) mechanism to select best routes for live video sequences in VANETs. MVIDE can be integrated with routing protocols to define routes considering the characteristics of multiple paths, vehicle mobility, and application requirements. This cross-layer mechanism ranks the quality of candidate paths based on the Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) method and, subsequently, assigns paths to different substreams. MVIDE was added to the GPSR-MA protocol, being called GPSR-MA-MVIDE. Simulating results show that GPSR-MA-MVIDE outperfoms GPSR-MA in terms of multi-flow handling and received video quality.*

1. Introduction

Vehicular Ad Hoc Network (VANET) is an emerging networking technology that uses moving vehicles to create a high mobility and wide range wireless network. Furthermore, with the rapid development of standards such as IEEE P1609 (WAVE), DSRC, and IEEE 802.11p, VANETs become a reality. As a result of that process, this technology has received a lot of attention in recent years and is targeted to support new services, including on-road multimedia safety and security and entertainment video flows [Felice et al. 2013].

Video stream supporting is an attractive service for VANET applications. Vehicles can cooperate among themselves to disseminate videos of dangerous situations allowing users and authorities (firefighters and paramedics) more meaningful information than scalar data, e.g., text messages [Piñol et al. 2012]. Nevertheless, transmission of such applications over VANETs still suffers from several open issues due to highly dynamic network topology and requirements of Quality of Experience (QoE) and Service (QoS) typical of multimedia applications. For instance, source, destination or relay vehicles can change direction abruptly, leading to a frequent and sometimes immediate disconnection.

To troubleshoot broken connection problems, multipath-based approaches can be employed as a promising alternative, since data path diversity is commonly abundant during peak periods of day in VANETs. Distribution of flows between two or more paths allows the network to achieve higher availability, load balancing, and resilience [Wang et al. 2012] [Mérindol et al. 2011]. In addition, a well-known technique to provide multistream coding suitable for transport over multiple paths is Multi-Description Coding (MDC) that creates two or more independent layers decodable in a stand-alone fashion [Wang et al. 2005]. MDC enables the traffic to explore the path diversity of the underlying VANET by sending alternative subtreams of the video along different routes. It guarantees better error resilience in cases of broken links, while increasing the packet delivery ratio and the human experience when watching a live video sequence [Qadri et al. 2012].

An issue that remains untouched in recent studies consists of the traffic distribution of network [Hashemi and Khorsandi 2012]. With the number of flows increasing, the probability of collision and buffer overflow in some vehicles will increase dramatically, and the network performance will degrade catastrophically [Zhuang and Ismail 2012]. Control schemes enable the network to distribute the flows between the nodes and thus avoiding these problems, as well as redirecting the traffic to other alternatives routes. The usage of underloaded nodes along the network allows end-users to receive better video quality. In addition, most of recent work have investigated only the route discovery process [Rezende et al. 2012] [Wahab et al. 2013], without considering different criteria concerning to network status, fair distribution and simultaneous multi-flows.

This paper proposes a cross-layer multi-criteria mechanism based on the vehicle mobility, current network conditions of each path, and application requirements to select multiple routes for live videos in VANETs, named Multi-flow-driven VIdeo DElivery (MVIDE). This mechanism can be integrated with a routing protocol and aims to improve the usage of scarce wireless resources, while increasing the human experience receiving video sequences. We design MVIDE as a Multiple-Criteria Decision-Making (MCDM) problem represented by the well-know Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) to conduct proper path ranking [Hwang and Yoon 1981]. TOPSIS is suitable for VANET scenarios by presenting high performance for multicriteria decisions in telecommunication and engineering environments [Naderi et al. 2013] [Taleb et al. 2012]. MVIDE divides each streaming into MDC substreams, which are forwarded to the best available routes. In case of overload in any node, MVIDE trigger the previous nodes to select a new path with available resource to keep higher video quality level from the user point-of-view. According to performance evaluation, MVIDE outperforms single-path approach and no multi-flow handling solution when number of flows increases. We measure the impact and benefits of MVIDE through simulation experiments with real video sequences in multi-path VANETs scenarios.

The remainder of this paper is organized as follows. Section 2 presents relevant related work. Section 3 introduces the MVIDE mechanism. Section 4 describes the coupling of MVIDE with the Greedy Perimeter Stateless Routing protocol with Movement Awareness (GPSR-MA). Simulation setup and evaluation results comparing GPSR-MA and GPSR-MA with MVIDE are presented in Section 5. Finally, conclusions are summarized in Section 6.

2. Related Work

The pervasiveness of mobile devices equipped with positioning systems (e.g., GPS) has significantly grown recently. In addition, high mobility of VANETs is shifting the attention of topology-centric to location-based protocols. Instead of their network addresses, geographic positions identify end-points of the communication [Felice et al. 2013]. Consequently, several work have investigated path discovery and dissemination of messages without concern about network status, multiple flows, and fair distribution of traffic in nodes [Xie et al. 2007].

Besides these work, in [Wang et al. 2012], authors extend the idea presented in [Rezende et al. 2012], developing a Location-aware Multipath Video Streaming scheme (LIAITHON) for video transmission in urban scenarios. This scheme uses positioning information, a route coupling prevention mechanism and a waiting time calculation to discover two relatively short paths with minimum route coupling effect. Although improvements regarding underlying single-path solution, the proposed scheme does not address the impact of distributing multiple video flows in VANETs. Therefore, coupling effects occur if there is more of one flow traversing the same communication range.

[Razzaq and Mehaoua 2010] suggest the use of Scalable Video Coding (SVC), multipath transmission, and a MCDM method called Grey Relational Analysis (GRA), to ranking and prioritizing important frames by transmitting them over the most suitable ranked routes. This work divides video stream to a base layer and enhanced layers then assigns path to different layers according to their importance aiming maximize throughput. With the usage of SVC scheme and GRA method, this approach operates well in term of packet delivery and overhead, however, a basic problem with SVC consists of complex interdependencies between the n-layers. Indeed, to decode a frame, the base layer must be obtained completely before reconstruction of enhancements layers [Xing and Cai 2012]. Thus, this work cannot guarantee the delivery of the intact base layer, which is necessary in SVC schemes. Furthermore, it does not ensure acceptable packet delivery time by using network coding at both source and intermediate nodes.

In urban scenarios, it will eventually lead to several streams being transmitted simultaneously. Once all the nodes in the network use the same routing strategy, a large amount of data will follow on more attractive nodes. Hence, it is vital to consider problems as higher congestion and loss ratio due to overhead on those nodes. [Hashemi and Khorsandi 2012] proposed a VANET Load Balanced Routing protocol (VLBR). It defines a traffic balancing between potential paths by attaining congestion feedback from the network and switching to lower congested routes by utilizing k-Shortest Paths algorithm. However, this work does not use video streaming and does not consider mobility and quality level parameters to path rankings.

In [Granelli et al. 2007] and [Granelli et al.], authors present the Greedy Perimeter Stateless Routing protocol with Movement Awareness (GPSR-MA), which from mobility parameters, retrieves information about the state of nodes in the network, such as speed, distance from destination, and direction. This proposal cover many aspects and takes into account several parameters, nevertheless, it is interesting and advantageous to consider not only mobility but also quality level parameters in the the path selecting process. In order to show the benefits of MVIDE, The GPSR-MA will be used as case study and underlying routing protocol in this work to apply MVIDE mechanism, as discussed in

Section 4. Therefore, our objective is to combine different independent criteria, including quality level parameters, into TOPSIS method for the path ranking. Further, the proposed mechanism also allows better distribution of the concurrent flows, when overload is detected in any node or route, hence the main contribution of this paper.

3. Multi-flow-driven VIdeo DElivery (MVIDE)

This section describes the proposed mechanism called MVIDE to enhance the control of simultaneous video flows in VANETs. The mechanism works jointly with an underlying routing geocast protocol and considers mobility and quality level attributes to transmit multiple video flows, while exploring the multipath capabilities of VANETs. Depending on the routing protocol, other different parameters can be incorporated aiming a more robust decision process. MVIDE occurs in three stages, namely Path Ranking, MDC Forwarding Decision Process, and Multi-flow Handling. The Path Ranking stage calculates and ranks the robustness and stability of all available paths. The MDC Forwarding Decision Process stage performs the forwarding process sending MDC pairs between chosen paths. The Multi-flow Handling stage obtains warnings from the monitor of traffic level, allowing dynamic change to other paths in case of congestion periods.

We consider the usage of video delivery on a VANET environment where there are k vehicles (nodes) moving over a grid area, each node has an identifier ($i \in [1, k]$), and the spatial distribution of nodes does not change very quickly in a short period of time. The combination of those nodes configures a dynamic graph $G(V, E)$, where vertices $V = \{v_1, v_2, ..., v_k\}$ mean a finite subset of k nodes, and edges $E = \{e_1, e_2, ..., e_m\}$ mean a finite set of asymmetric wireless links between them. We denote a subset $N(v_i) \subset V$ as all 1-hop neighbors within the radio range of a given node v_i. Further, each node v_i has a queue (Q) with a maximum queue capacity (Q_{max}) and current queue length (Q_{length}). The queue policy schedules the packet transmission by using First In First Out (FIFO) and drops packets by using Drop Tail in case of buffer overflow. Each node v_i has a positioning system, such as GPS, so that it is aware of location and accurately synchronized in time.

Through positioning and traffic periodic information exchanged between a subset $N(v_i)$, each node v_i is able to estimate the follows parameters: 1-hop distance (d), direction (θ), speed (s), delay (τ), and the buffer level (ϕ) of its neighbors. Thus, each node v_i is aware of the latest state $S(d, \theta, s, \tau, \phi)$ of its neighborhood. For a given $G(V, E)$, which represents a multi-hop and multi-flow network, there is a set of real-time video flows $F = \{f_1, f_2, ..., f_n\} | n \leq \frac{k}{2}$. Furthermore, this network is composed of a finite subset of source nodes $VS = \{vs_1, vs_2, ..., vs_n\}$ and a finite subset of destination nodes $VD = \{vd_1, vd_2, ..., vd_n\}$, where $\forall\, vs_j \in VS \; \exists!\, vd_j \in VD | (j \in [1, n])$. Thus, for each vs_j there is a respective vd_j resulting in n source-destination pairs $(VS, VD) \subset V$, and the vd_j position is known. Each stream f_j has different sizes and characteristics and its transmission from vs_j to its respective vd_j demands a transmission request from vd_j composed of the vsj identifier (logical address), its location (x_{vs_j}, y_{vs_j}) and the f_j identifier.

The node state $S(d, \theta, s, \tau, \phi)$ enables MVIDE to define and select different possible paths, taking into account the topology dynamism to adjust each flow for increasing the link resilience. Furthermore, MVIDE redirects flows to underloaded paths in case of congestion likelihoods. Thus, it supports robust real-time video distribution and, consequently, a better QoE support for video streaming in multi-flow scenarios. We will present each stage in detail in the following subsections.

3.1. Path Ranking

The Path Ranking stage allows MVIDE to calculate, among available paths, routes with better link quality and stability. MVIDE considers performance metrics and location features $S(d, \theta, s, \tau, \phi)$, previously defined, for obtaining the state information of each node, as factors in the TOPSIS method. It consists of a multi-criteria method to classify among a set of alternatives the best one, so that the chosen alternative should not only have the shortest distance from 'positive ideal' reference point (A^+), but also the farthest distance from a 'negative ideal' reference point (A^-). A^+ has the most benefits and lowest cost of all alternatives, and A^- is the one with the lowest benefits and highest cost [Hwang and Yoon 1981]. In MVIDE, the TOPSIS alternatives $A = \{A_1, A_1, ..., A_m\}$ mean all paths in which it is possible to calculate their respective factors $n = 5$, thus the number of alternatives m varies according to the available paths.

The TOPSIS operation is expressed in a series of steps [Naderi et al. 2013]. Thus, given a source-destination pair $(vs_j, vd_j) \in V$ and a flow f_j, respectively, MVIDE firstly selects paths for the flow f_j. Each flow f_j contains a subset of forwarding nodes $VF_j = \{vf_{j,1}, vf_{j,2}, ..., vf_{j,z}\} \subset V$ that establishes routing paths from vs_j to vd_j and transports a subtream MDC originated from f_j. Thereby, it is built the decision matrix D (Eq. (1)), where x_{pq} describes the value of alternative $A_p|(p \in [1, m])$ and factor $C_q|(q \in [1, n])$.

$$D = \begin{array}{c c} & \begin{array}{cccc} C_1 & C_2 & \cdots & C_n \end{array} \\ \begin{array}{c} A_1 \\ A_2 \\ \vdots \\ A_m \end{array} & \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1} & x_{m2} & \cdots & x_{mn} \end{bmatrix} \end{array} \tag{1}$$

MVIDE selects only one path for each MDC substream, as shown in Subsection 3.2. Thus, for each ranking, the best disjoint paths are chosen from matrix D by using linear combination. In addition, the raw data are normalized to eliminate differences by measurement units and scales, similarly to several MCDM problems. Decision matrix R receives the normalized elements of matrix D by using Eq. (2):

$$r_{pq} = \frac{x_{pq}}{\sqrt{\sum_{p=1}^{n} (x_{pq})^2}} \quad p = 1, 2, ..., m \quad and \quad q = 1, 2, ..., n \tag{2}$$

At this point, considering the different weight ω_q for each factor, the weighted normalized decision matrix W is calculated by multiplying the values r_{pq} in R and the importance weights of evaluation criteria. As result, W is defined as Eq. (3):

$$W = [w_{pq}]_{m \times n} | w_{pq} = r_{pq} \times \omega_q, \quad p = 1, 2, ..., m \quad and \quad q = 1, 2, ..., n \tag{3}$$

Using the measures in the matrix W, A^+ and A^- are identified by all best and worst performance scores (Eqs. (4) and (5)), respectively.

$$A^+ = \left\{w_1^+, \cdots, w_n^+\right\} = \left\{(Maxw_{pq} \mid q \in Q), (Minw_{pq} \mid q \in Q')\right\} \tag{4}$$

$$A^{-} = \left\{w_1^{-}, \cdots, w_n^{-}\right\} = \left\{(Minw_{pq} \mid q \in Q), (Maxw_{pq} \mid q \in Q')\right\} \quad (5)$$

For MVIDE, all factors $S(d, \theta, s, \tau, \phi)$ are considered as cost factors. This allows the definition of A^+ and A^-, and the identification of distances (d_p^+ and d_p^-) ,i.e., the relative proximity between alternatives. Thus, the distances of A_p to A^+ and A^- are calculated by using Eqs. (6) and (7), respectively:

$$d_p^{+} = \sqrt{\sum_{q=1}^{n} \left(w_{pq} - w_q^{+}\right)^2}, \quad p = 1, 2, ..., m \quad and \quad q = 1, 2, ..., n \quad (6)$$

$$d_p^{-} = \sqrt{\sum_{q=1}^{n} \left(w_{pq} - w_q^{-}\right)^2}, \quad p = 1, 2, ..., m \quad and \quad q = 1, 2, ..., n \quad (7)$$

Since the closeness relative is computed, the ranking order of all alternatives is obtained, allowing selecting the most feasible paths. Eq. (8) expresses the closeness coefficient S_p of each alternative:

$$S_p = \frac{d_p^{-}}{d_p^{+} + d_p^{-}}, \quad p = 1, 2, ..., m \quad (8)$$

Lastly, the index value of S_p varies between 0 and 1 and all alternatives are ordered in a descending way. Under these circumstances high S_p means lower distance to A^+ while lower S_p establishes great distance to the best solution. From use of these values for ranking all possible paths as alternatives, it is possible to obtain more suitable paths of the list.

3.2. MDC Forwarding Decision Process

This stage employs the forwarding task of the live video flows along the routes chosen in the Path Ranking stage. In this stage, given a source-destination pair (vs_j, vd_j), the Multi-Stream Video Encoder component (from MDC architecture) divides a flow $f_j \in (VS_j, VDj)$ in z independet substreams, called MDC descriptors $MD_j^c|(c \in [1, z])$. Each MD_j will be transported by the routing protocol along z set of disjoint paths established each one by forwarding nodes.

For simplification, MVIDE will be analysed only with two substreams (MDC descriptors), i.e., $z = 2$ (MD_j^1 and MD_j^2). Figure 1 illustrates the architecture of the system combining MDC, multipath, and MVIDE. To start a video transmission, vd_j requests a video from vs_j, thus $S(d, \theta, s, \tau, \phi)$ is exchanged periodically among $N(vs_j)$ and sent to vs_j. The information update periods follow a spatial distribution of nodes whose variation does not change rapidly in a short period of time.

Once vs_j divides a multimedia stream into MD_j^1 and MD_j^2, the Traffic Allocator distributes the packets from these substreams among the two best paths. Then, the routing protocol will route the substreams along the two paths. At each update period, the routing protocol detects the available paths between the vs_j and vd_j and informs to vs_j.

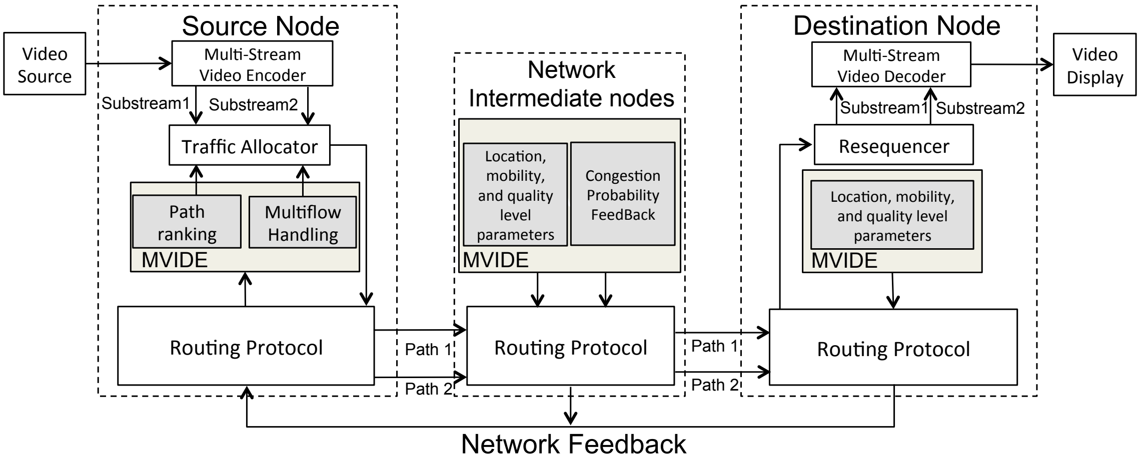


Figure 1. MDC architecture with MVIDE.

In the Multi-flow Handling phase, vs_j receives information about the congestion likelihood of routes in activity, as detailed in the next subsection. At vd_j, the Resequencer component retrieves each packet arriving from the two paths and reassembles it after a predetermined time period. At last, Multi-Stream Video Decoder reconstructs the video from the received packets in each substream. The Multi-Stream Video Encoder and Decoder, as well as Traffic Allocator and Resequencer components, belong to the MDC architecture and do not overload the system [Wang et al. 2005].

MVIDE filters the Path Ranking through a *max delay* and a *forwarding zone* threshold to decrease the processing time and memory consumption. If there is no more than one available path, i.e., channel conditions cannot assure the video requirements, it is preferable that only the available path forwards both substreams [Zhuang and Ismail 2012], thus MVIDE redirects the substreams as single path strategy.

3.3. Multi-flow Handling

This stage controls the multi-flow handling on intermediate nodes. Figure 2 shows a logical representation of the progressive download task in a forwarding node vf_j, where multiple flows are received simultaneously. With the number of flows increasing, intermediate nodes can receive heavy traffic, which indicates a poor distribution of flows in the network. This can cause multiplexing, buffer overflow, delay, packet loss, leading to degradation of the video quality perceived by the user.

When the buffer of an intermediate node starts to be full, this node should not receive more flows, as shown in Figure 2. Thus, each vf_j should also know the congestion probability of its neighbors. To establish better flow distribution, we define the buffer level ϕ_j as the number of packets stored in the buffer queue of a vf_j. Furthermore, each forwarding node vf has knowledge about the current amount of data traversing it, i.e., the packets per second that it is receiving and transmitting $trafficLevel = rateOut - rateIn$. Hence, a given vf_j is able to calculate its congestion indicator CI_j according to Eq. (9).

$$CI_j = \frac{1}{1 + e^{(-c(trafficLevel_j) - Q_{max}/2)}} \tag{9}$$

We employed the sigmoid function for the exponential distribution of CI. The

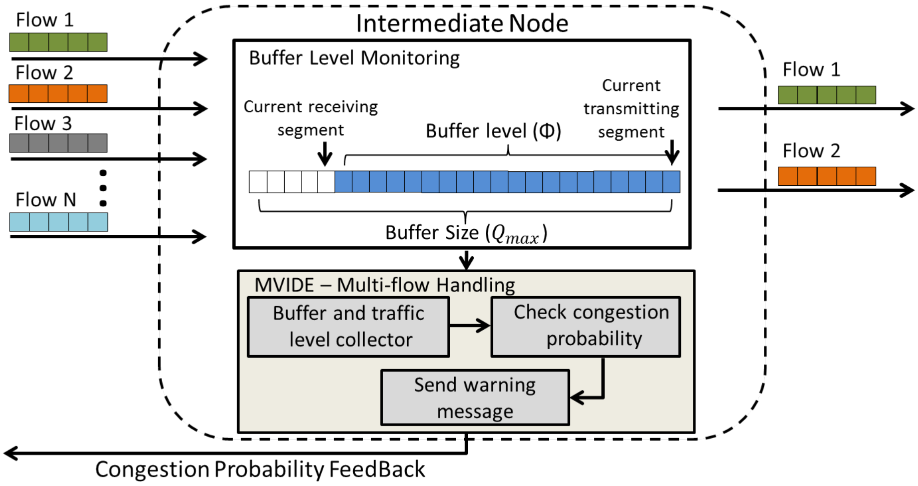


Figure 2. MVIDE into an intermediate node.

parameter $Q_{max}/2$ determines the center of the sigmoid output, and c regulates the slope or "growth rate" of this function during its rising portion. The parameter c has a negative value to enable large values of $trafficLevel$ and to generate an output closer to 0. Finally, the congestion probability $\rho_j(CI_j, \phi_j)$ of a given node vf_j increases when $trafficLevel_j$ rises rapidly together with high values of ϕ_j. It is calculated in the weighted Eq. (10):

$$\rho_j^t(CI_j, \phi_j) = \alpha \cdot CI_j^t + (1 - \alpha) \cdot (\phi_j^t) \tag{10}$$

Where $\rho_j^t(CI_j, \phi_j)$ is a congestion probability at a given instant t on a vf_j and α is a weighted factor. Hence, the greater the amount of flow traversing vf nodes, the longer $\rho(CI, \phi)$. This means that MVIDE not only allows the handling of the traffic load distribution, but also deals with the transmission latency.

Through Eq. (10), we attempt to exploit, when a given vf_j experiences many concurrent streams. Thus, vf_j uses its $\rho(CI, \phi)$ to notify the previous nodes through a greedy message. The network feedback collaborates to make decisions aiming QoE improvements. Inspired by [Hashemi and Khorsandi 2012], we opted by readjusting a new route for the traffic allocator when a predefined congestion threshold is exceeded. Based on the path ranking, a new alternative route is selected to replace the paths without quality enough for the live video flows. This new route will be used until either the arrival of the new warning message or the selection of a new best route by path ranking scheme.

4. Integration of MVIDE with the GPSR-MA protocol

This section presents the MVIDE mechanism in addition to the underlying routing protocol. To assess MVIDE functionalities, we develop and adapt its three stages jointly with the geocast protocol GPSR-MA [Granelli et al. 2007] [Granelli et al.]. GPSR-MA improves the well-know GPSR protocol by using not only parameters related to distance, but also some of the factors, which allows MVIDE to select the best routes, such as speed and direction information.

In VANET scenarios, vehicles at a given instant t move along well defined segments, which form the roads. Each node performs neighbor position estimates at time

$t + 1$ by using the above mentioned parameters (distance, speed, and direction). GPSR-MA creates three functions to calculate these parameters in the routing process: $g(d)$, $f(s)$, and $h(\theta)$, given by Eqs. (11a), (11b) and (11c), respectively.

$$g(d) = e^{-(t_{range}+d-d_i)} \qquad f(s) = e^{-(\frac{(x-s_i)^2}{2\sigma^2})} \tag{11a,11b}$$

$$h(\theta) = \begin{cases} 0 & if \quad \theta \geq \frac{\pi}{2} \\ cos^2(\theta) & elsewhere \end{cases} \tag{11c}$$

Where t_{range} means transmission range and d_i is the distance of current node i from destination. GPSR-MA employs a Gaussian function to a given node v_i calculate the difference between the speed x of $N(v_i)$ in comparison with its reference speed s_i within a predefined standard deviation σ. To calculate movement direction component, θ means the angle between node movement direction and the line connecting it to destination.

The MVIDE mechanism coupled with GPSR-MA protocol, called GPSR-MA-MVIDE, uses other link quality factors (namely delay and traffic level) allowing a robust Path Ranking in addition to the position-based parameters of GPSR-MA. To add the MVIDE functionality with GPSR-MA protocol, we establish criteria as input to the MCDM method without affecting the protocol performance. As defined above in Eqs. (11a), (11b), and (11c), GPSR-MA periodically exchanges messages containing location and mobility information. Thus, GPSR-MA-MVIDE also uses delay and traffic level information to determine the two best route options.

The arrival-time between messages suggests delay measurements. This information is important, since some vehicles with good mobility and position parameters may have higher transmission delay due to areas with higher probability of collisions, i.e., nodes with larger contention windows. As the number of transmissions increases, there will be more contention on the channel and back-off values will be higher. Hence, we add the delay transmission values of signaling messages for the MVIDE Path Ranking stage.

The traffic level determines whether forwarding nodes face heavy load, which may impair its flow transmissions. Thus, the buffer level ϕ allows MVIDE to determine the traffic level of candidate nodes. Once established the best paths, the forwarding nodes transmit the congestion probability $\rho(CI, \phi)$ only if it exceeds a given threshold. The buffer level and delay information allows source node to run a more robust path ranking.

Figure 3 illustrates the path selection process to GPSR-MA-MVIDE protocol (represented by ***ii***) in comparison to the pure GPSR-MA protocol using MDC and multipath (GPSR-MA-MDC, represented by ***i***). For a given time t there is only one flow in the network from S_1 to D_1. In this case, both GPSR-MA-MDC and GPSR-MA-MVIDE have the same choices for paths, i.e., they select the paths with shortest distance to the destination node. However, for a time $t+1$, a new stream starts from $S2$ to D_2 and, through Path Rankin stage, GPSR-MA-MVIDE selects an alternative route (S_3) for a MDC substream instead of the path with shortest distance to the destination node, in this way it relieves vf_3 that already transmits data. The same applies for time $t + 2$, where, through Multi-Flow-Handling stage, MVIDE achieves a better distribution of flows in the network. On the other hand, GPSR-MA-MDC concentrates all traffic in more attractive nodes, such as vf_3, which can impair the concurrent flows.

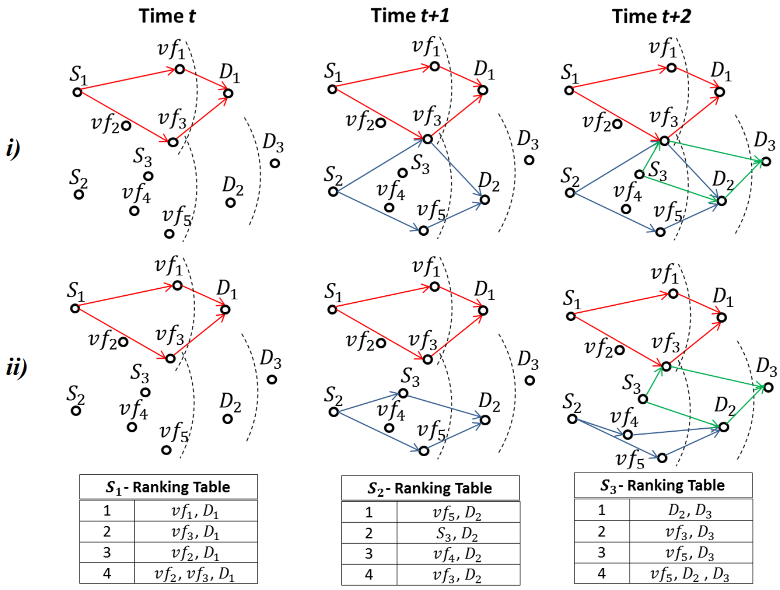


Figure 3. *i)* GPSR-MA-MDC and *ii)* GPSR-MA-MVIDE.

Other protocols can implement MVIDE. It might be suitable for protocols that use SNR level rather than distance [Felice et al. 2013], as well as added in multipath schemes. Depending on the routing strategy, the MVIDE steps can be easily adapted.

5. Performance Evaluation

We employed the Network Simulator version 2.33 (NS2) to perform the simulations and SUMO to generate the vehicular movement traces. The mobility model is the Manhattan Model. More details about the simulation parameters are shown in Table I. For simplification, in this work MVIDE produces only two substreams for each video flow and considers an area large enough to cover several live videos flows with two paths each one.

Table 1. Simulation Parameters.

Parameter	Value
Area Size	1600m x 1600m
Average road length	400m
Speed Range	5.5 to 15.5 m/s
Beacon interval	0.3s
Radio Range	250
Number of Vehicles	60 to 220
Simulation Time	500s
MAC Layer	IEEE 802.11p
Queue Capacity	30 pkts
Propagation Model	Nakagami Dist.

Figure 4. Video sequences.

In order to establish a relevant scenario where multiple video streams with different sizes and characteristics (bitrate and length) are requested and generated as in a real scenario, we have used EvalVid - A Video Quality Evaluation Tool-set. This framework allows evaluating the video streaming quality and getting more realistic results. In this way, we have conducted the simulation by transmitting 15 real and widely used different MPEG sequences of natural scenes (352 x 288 pixels) with duration varying from 5s to 70s and internal GoP structure configured as two B-frames for each P-frame (IBBP pattern) [Seeling and Reisslein 2012]. Figure 4 shows one frame of each video.

To demonstrante the impact of MVIDE, we use the single path protocol, GPSR-MA, for comparison. Also, to investigate the performance gain when taking into account the handling of multi-flows caused by overhead, we have compared GPSR-MA-MVIDE with a GPSR-MA multipath-MDC version (GPSR-MA-MDC), i.e., this protocol sends one MDC substream to two different paths. Thus, the lost frames from one substream are compensated by those from the other substream, whereas lost frames from both paths cannot be reconstructed. In that case, the video decoder performs the Frame-Copy error concealment technique, which uses the last well-received frame to replace lost frames.

The impact and benefits of the proposed solutions were measured by the following metrics: Packet Delivery Rate (PDR), average number of hops per source-destination pair, and End-to-end Delay. From the received video quality point-of-view, measurements were carried out with Structural SIMilarity (SSIM). It is a well-known QoE metric, which measures the structural distortion of the video to obtain a better correlation with the user's subjective impression. These metrics determine the behavior of GPSR-MA-MVIDE regarding the GPSR-MA-MDC and pure GPSR-MA. It also presents the results with the variations in number of connections (flows), number of vehicles, overhead generated, and the quality of routes for the video delivery.

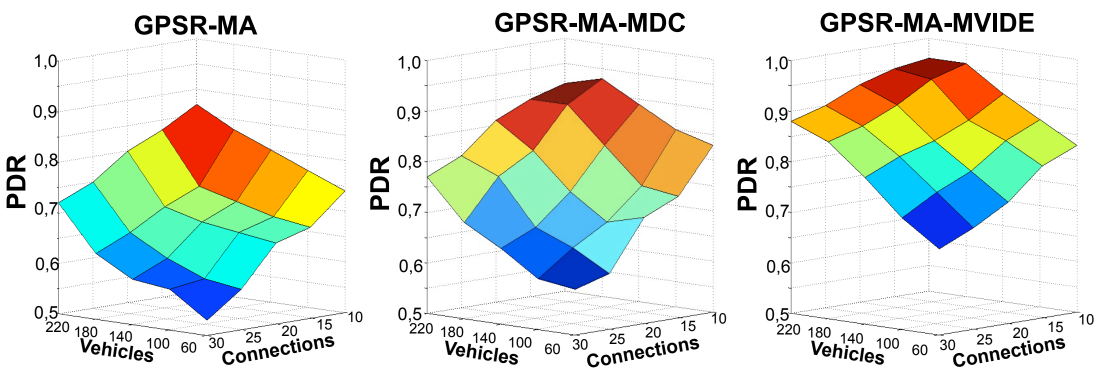
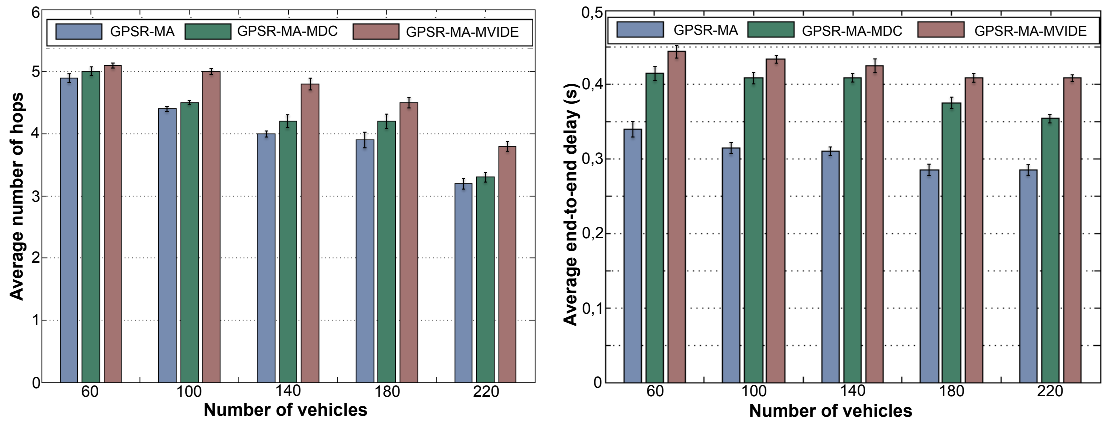


Figure 5. PDR performance comparison among GPSR-MA, GPSR-MA-MDC, and GPSR-MA-MVIDE when number of vehicles and connections (flows) varies.

In terms of network performance and PDR, Figure 5 shows the performance results for the three above listed cases: GPSR-MA protocol, GPSR-MA with MDC and multipath (GPSR-MA-MDC), and GPSR-MA coupled with MVIDE mechanism (GPSR-MA-MVIDE). Each plotted result in the graphs is the average of results generated from 35 different simulations. The confidence interval was calculated with 95% confidence level. As we can observe from the simulation results, GPSR-MA-MVIDE notably outperforms other solutions when number of vehicles and connections increases. The major rationale for this behavior is that GPSR-MA faces several broken link situations when there are few available vehicles. Whereas, GPSR-MA-MDC and GPSR-MA-MVIDE neatly distribute the load into two complementary paths, which reduces the broken link problem of GPSR-MA. Despite being a multipath solution, GPSR-MA-MDC does not gain any performance improvement facing multiple flows. The problem with this solution is the increasing number of concurrent flows in the network. As all the nodes use the same routing strategy, many streams are directed to the same more attractive paths. This can cause congestion, overload, and collision in some nodes, consequently, lead to degradation in received video qualities. However by using the Path Ranking and Multi-flow Handling stages based on traffic level, GPSR-MA-MVIDE ensures better traffic distribution of the network, resulting in more PDR.



(a) (b)

Figure 6. (a) Number of hops comparison. (b) End-to-end delay comparison.

Figure 6a presents the average number of hops comparison per source-destination pair among the discussed approaches. This figure shows that the paths for the MDC substreams (GPSR-MA-MDC and GPSR-MA-MVIDE) exceeds ones of the single-path solution (GPSR-MA), which follows the same pattern as there are more paths and more probability over which packets can be forwarded in an MDC scheme. Moreover, due to Path Ranking, GPSR-MA-MVIDE considers other quality level metrics to classify routes. Thus, it is not restricted to shortest paths and mobility parameters such as distance.

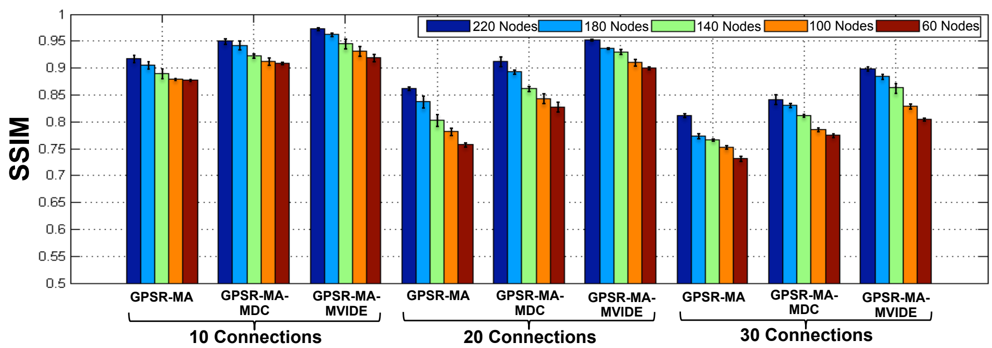


Figure 7. Average video quality evaluation.

In the last measure we analyze the average end-to-end delay of each frame, which is slightly lower mainly in GPSR-MA compared to GPSR-MA-MVIDE (Figure 6b). As mentioned previously, in MVIDE, according to its operating model, provides more effort to deliver flows when faces congestion, this could mean forwarding of substreams to alternatives sparse roads with increasing transmission durations. On the other hand, the other two protocols just drop packets in these situations and will not make any effort. This might result in longer delays and path lengths. However, delay levels are less than one second, which are negligible even in video application and are significantly smaller than requirements of 4 to 5 seconds defined by CISCO [Hatting et al. 2005].

To demonstrante the impact of the proposed solution on the human perception, the results in Figure 7 present the SSIM metric. With 10, 20 and 30 connections, MVIDE keeps the video quality high, i.e., SSIM around 0.94, 0.93, and 0.86, respectively. An average increase of 15% and 9% compared to GPSR-MA and GPSR-MA-MDC. It confirm the results obtained in the Figure 5 and show similar benefits to those achieved earlier. SSIM values range from 0 to 1, where a higher value means better video quality. As we can see, in scenarios with 10 connections, the three protocols maintain a similar video quality level. However, in scenarios with 30 connections at the same time, the difference between them becomes very significant.

From the graphs aforementioned, is clear that GPSR-MA-MVIDE preserves more video quality because of its path ranking and multi-flow handling procedure in which it selects and chooses the best options as next hops and switches to other routes as soon as congestion is identified. As result, in the performance network, GPSR-MA-MVIDE will also outperform GPSR-MA and GPSR-MA-MDC in congested periods.

6. Conclusions

This paper introduced MVIDE to enable an efficient on-road real-time multiple videos dissemination with QoE support in VANETs. MVIDE calculates and ranks the best routes to distribute simultaneous MDC substreams in the network. It was integrated with GPSR-MA protocol and considers the characteristics of multiple paths, vehicle mobility, and application requirements. Further, MVIDE allows multi-flow handling by using information about traffic level in intermediate nodes to adjust paths without breaking connections, providing a better network traffic distribution. Simulation results highlight the performance, effectiveness and QoE support of MVIDE by measuring the video quality levels when the number of vehicles and flows varies. We measured the video quality of each transmitted video by means of QoS and QoE metrics. MVIDE provided multimedia dissemination with robustness and QoE support compared to single-path GPSR-MA protocol and GPSR-MA with MDC scheme.

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